

A real-axis spectral dictionary for the arithmetic of number fields

Aaron Jones

JtechAI LLC

June 15, 2026

Abstract

The Dedekind zeta function ζ_K of a number field K , read as the grand partition function of a “ K -primon gas” (a Gibbs family $p_{\mathfrak{n}} \propto N\mathfrak{n}^{-\beta}$ over integral ideals), has a Fisher information $I_K(\beta) = \partial_{\beta}^2 \log \zeta_K(\beta)$ whose expansion about the Hagedorn point $\beta = 1$ is governed by the nearest singularity of ζ_K ; under a per-field hypothesis $\text{ZFD}(K)$ this is a trivial zero on the real axis. We show that this single expansion simultaneously encodes the leading arithmetic of K : its *unit rank* $\rho = r_1 + r_2 - 1$ (from the radius of convergence and a distance-aware amplitude law), its *signature* (r_1, r_2) (from the orders of the next trivial zeros together with the degree), the product hR/w of class number, regulator, and roots of unity (from a sub-leading log-singularity amplitude), and the *discriminant* $|d_K|$ (from the constant term). Each reading is a corollary of classical facts—the order of ζ_K at $s = 0$, the class-number formula, and the residue at $s = 1$ —so the contribution is not new arithmetic but a *reading*: the real-axis thermodynamics of the prime gas carries this data with no analytic continuation into the complex plane and no use of zero locations beyond $\text{ZFD}(K)$. We prove the rank-reading law (Lemma 1) and exhibit it numerically for $\mathbb{Q}$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{2})$, a cyclic cubic field, and the mixed cubic $\mathbb{Q}(\sqrt[3]{2})$ (the last requiring a non-abelian Artin L -function). The general statement is conditional on $\text{ZFD}(K)$ excluding a Siegel-type zero inside the relevant disk; we are explicit about this wall.

1 Introduction and scope

Reading $\zeta(s) = \sum_n n^{-s}$ as the grand partition function of a gas of “primons”—bosonic quanta labelled by primes, with energies $\log p$ —is by now classical [1, 2, 3]. In this picture the temperature axis $\beta = \Re s > 1$ is physical and the Hagedorn point $\beta = 1$ is a limiting-temperature singularity. The object of this paper is the *Fisher information* of the Gibbs family, $I_K(\beta) = \text{Var}_{\beta}(\log N\mathfrak{n}) = \partial_{\beta}^2 \log \zeta_K(\beta)$, for the generalization in which ζ is replaced by the Dedekind zeta function ζ_K of a number field K (the K -primon gas, with states the integral ideals $\mathfrak{n}$ weighted by $N\mathfrak{n}^{-\beta}$).

Our claim is a *dictionary*: the Taylor expansion of I_K about the Hagedorn point reads off the leading arithmetic invariants of K . The mechanism is elementary—the expansion is singular at the zeros and pole of ζ_K shifted to the origin, so its radius and amplitudes are determined by the nearest such singularities—but the reading is, to our knowledge, not assembled elsewhere, and it has two features worth stating at the outset.

Real axis, no continuation. Every quantity below is computed from $\zeta_K(\beta)$ for real β near 1. We never continue into the critical strip; the nontrivial zeros enter only as the abstract “next singularity” whose *distance* bounds a radius of convergence, never as objects whose location is asserted or used. **NO-RH-CLAIM:** nothing here bears on the location of any zero of any L -function.

Contribution class: framing. The arithmetic inputs— $\text{ord}_{s=0} \zeta_K = \rho$, the analytic class-number formula $\zeta_K^*(0) = -hR/w$, and $\text{Res}_{s=1} \zeta_K = 2^{r_1}(2\pi)^{r_2}hR/(w\sqrt{|d_K|})$ [4, 5]—are all classical. The contribution is that one real-axis Fisher expansion encodes signature, hR/w , and discriminant *simultaneously*. We label this honestly as a framing-class result, not a new theorem of arithmetic.

2 The Fisher expansion and the dictionary

Write $\varepsilon = \beta - 1$ and $G(\varepsilon) = \log [\varepsilon \zeta_K(1 + \varepsilon)]$. Since $(s - 1)\zeta_K(s)$ is entire of order one (Hecke), G is analytic at $\varepsilon = 0$; let $b_k = [\varepsilon^k]G$. The Fisher information is $I_K(1 + \varepsilon) = \partial_\varepsilon^2 \log \zeta_K(1 + \varepsilon) = \varepsilon^{-2} + \partial_\varepsilon^2 G(\varepsilon)$, so the regularized part G carries all field-dependent information. The four readings (Table 1) follow from the singularity structure of ζ_K on the real axis.

invariant	read from	classical mechanism	status
unit rank $\rho = r_1 + r_2 - 1$	radius of $\{b_k\}$ / amplitude $ b_k k R_c^k \rightarrow m$	$\text{ord}_{s=0} \zeta_K = \rho$	[V] tested; [V] ($\mathbb{Q}(\sqrt[3]{2})$)
signature (r_1, r_2)	orders at distances 2, 3 (give $r_2, r_1 + r_2$)	$\text{ord}_{s=-(2j-1)} \zeta_K = r_2$	[V], 13 digits
hR/w	sub-leading log-amplitude $A = e^{H(-1)}$	$\zeta_K^*(0) = -hR/w$	[V], exact
$ d_K $	constant term $\div$ amplitude, Res/A	residue formula at $s = 1$	[V], exact

Table 1: The dictionary. $H(\varepsilon) = G(\varepsilon) - m \log(1 + \varepsilon)$ removes the binding-zero contribution of order m ; $A = e^{H(-1)}$. All four are read off one expansion about $\beta = 1$.

3 The rank-reading law

The trivial-zero structure of ζ_K is fixed by the signature through the Γ -factors of the completed function $\Lambda_K(s) = |d_K|^{s/2} \Gamma_{\mathbb{R}}(s)^{r_1} \Gamma_{\mathbb{C}}(s)^{r_2} \zeta_K(s)$ (with $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2} \Gamma(s/2)$, $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s} \Gamma(s)$). Λ_K is meromorphic with simple poles only at $s = 0$ and $s = 1$, and $\Lambda_K(s) = \Lambda_K(1 - s)$ [4]. Since $\Gamma_{\mathbb{R}}$ and $\Gamma_{\mathbb{C}}$ together contribute poles of total order $r_1 + r_2$ at $s = 0$ while Λ_K has only a simple pole there, ζ_K must vanish to order $(r_1 + r_2) - 1 = \rho$; likewise (no Λ_K pole at negative integers) $\text{ord}_{s=-(2j-1)} \zeta_K = r_2$ and $\text{ord}_{s=-2j} \zeta_K = r_1 + r_2$ for $j \geq 1$. The nearest zero of ζ_K to $s = 1$ thus sits at distance 1 (the $s = 0$ zero, order ρ) when $\rho \geq 1$, and at distance 2 (resp. 3) for an imaginary quadratic field (resp. $\mathbb{Q}$).

Lemma 1 (Rank-reading law). *Let R_c be the radius of convergence of $\{b_k\}$ (we reserve R for the regulator) and m the order of the binding zero at distance R_c . Assume ZFD(K): ζ_K has no nontrivial zero in the punctured disk $0 < |s - 1| < R_{\text{triv}}$, where $R_{\text{triv}} \in \{1, 2, 3\}$ is the distance to the nearest trivial zero (1 if $\rho \geq 1$; 2 for imaginary quadratic; 3 for $\mathbb{Q}$). Then*

$$|b_k| k R_c^k \xrightarrow{k \rightarrow \infty} m, \quad R_c = R_{\text{triv}}. \quad (1)$$

In particular $\rho \geq 1 \Rightarrow R_c = 1$ and $|b_k| k \rightarrow m = \rho$: the expansion reads the exact unit rank. The spine is unconditional: $\rho \geq 1 \Rightarrow R_c \leq 1$, equivalently $R_c \geq 2 \Rightarrow \rho = 0$; the equality $R_c = R_{\text{triv}}$ is what requires ZFD(K).

Proof sketch. The Hadamard factorization of the order-one entire function $(s - 1)\zeta_K(s)$ writes $G(\varepsilon)$ as a sum of terms $m_0 \log(1 - \varepsilon/\varepsilon_0)$ over the zeros $\varepsilon_0 = \rho_0 - 1$ of ζ_K (the $(s - 1)$ factor removes the pole), with Taylor coefficients $-m_0/(k \varepsilon_0^k)$. The radius R_c is the distance to the nearest ε_0 . Under ZFD(K) that nearest singularity is the *real* trivial zero—for $\rho \geq 1$ the $s = 0$ zero, at $\varepsilon_0 = -1$ (whence the $H(\varepsilon) = G(\varepsilon) - m \log(1 + \varepsilon)$ of Table 1)—so a single real term dominates and $|b_k| k R_c^k \rightarrow m$. Were the nearest zeros a non-real conjugate pair (the low-lying-zero scenario ZFD(K) excludes) the coefficients would oscillate and only $\limsup = m$ would

survive; the convergence of $|b_k| k R_c^k$ is thus itself a (weak) consequence of $\text{ZFD}(K)$. Full proof: [7, T1lemma_E1_proof_v0.2]. $\square$

Remark 1 (The Siegel wall, stated plainly). $\text{ZFD}(K)$ is *not* implied by GRH: a zero at $\frac{1}{2} + it$ lies inside the disk $|s - 1| < 1$ whenever $|t| < \sqrt{3}/2 \approx 0.866$. If $\text{ZFD}(K)$ fails through a Siegel real zero $\beta_0 \in (0, 1)$, the binding singularity moves to distance $1 - \beta_0 < 1$: the amplitude law still returns the multiplicity ($|b_k| k R_c^k \rightarrow 1$, the Siegel zero being simple), but the *radius* is misread as $1 - \beta_0$ rather than 1, so the inference $R_c = 1 \Rightarrow \rho \geq 1$ silently fails—most insidiously for a rank-1 field, where the amplitude is also 1 and only the shrunken radius betrays the contamination. The general rank-reading is therefore $[E]$, conditional on $\text{ZFD}(K)$ per field; an unconditional statement would require a no-Siegel-zero theorem plus a uniform low-lying-zero bound, neither available. We do not move this wall.

4 Verified instances

For each field we compute $\{b_k\}$ to high precision and read $|b_k| k R_c^k \rightarrow m$ (to 6–13 significant figures), cross-checking the order by $\zeta_K(2\varepsilon)/\zeta_K(\varepsilon) \rightarrow 2^m$. For the three fields of rank ≥ 1 we verify $\text{ZFD}(K)$ directly by an argument-principle integral $(2\pi i)^{-1} \oint_{|s-1|=r} \zeta'_K/\zeta_K ds$, which counts (zeros – poles) enclosed. At $r = 0.90$ (and, for $\mathbb{Q}(\sqrt{5})$, refined to 10^{-12} at $r = 0.98$) it returns -1 —the simple pole at $s = 1$, with *no* enclosed zero—certifying $\text{ZFD}(K)$ on $0 < |s-1| < r$. The binding trivial zero at distance 1 lies just outside these contours and is detected instead by the b_k tail, which pins $R_c = 1$ to seven figures; the two methods bracket the hypothesis.

field K	signature	rank ρ	(R_c, m) read	route
$\mathbb{Q}$	(1, 0)	0	$R_c = 3, m = 1$	ζ
$\mathbb{Q}(\sqrt{-3})$	(0, 1)	0	$R_c = 2, m = 1$	$\zeta \cdot L(\chi_{-3})$
$\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{2})$	(2, 0)	1	$R_c = 1, m \rightarrow 1$	$\zeta \cdot L(\chi)$, Hurwitz
cyclic cubic, cond. 7	(3, 0)	2	$R_c = 1, m \rightarrow 2$	$\zeta \cdot L(\chi)L(\chi^2)$
$\mathbb{Q}(\sqrt[3]{2})$	(1, 1)	1	$R_c = 1, m \rightarrow 1$	degree-2 S_3 Artin L (AFE)

Table 2: Rank-reading verified across abelian and non-abelian routes.

The mixed cubic $\mathbb{Q}(\sqrt[3]{2})$ is the most stringent instance: signature (1, 1), so ζ_K factors as $\zeta \cdot L(\sigma, \cdot)$ with σ the non-abelian degree-two Artin representation of the S_3 Galois closure, outside the Dirichlet/Hurwitz route. We construct $L(\sigma)$ from $d_K = -108$, resolvent $\mathbb{Q}(\sqrt{-3})$, conductor $\mathfrak{f} = (6)$ with $N\mathfrak{f} = 36$ (conductor–discriminant), and the cubic residue symbol as the prime seed (verified at all 20 primes below 80 against root counts of $x^3 - 2 \bmod p$). The Artin root number is $W(\sigma) = +1$, whence $W(\zeta_K) = W(\zeta)W(\sigma) = +1$ (Hecke); its approximate functional equation is validated by the theta/functional-equation identity to 10^{-31} , with a wrong-root-number control failing by $O(1)$. Result: $|b_k| k \rightarrow 1$ (1.00012507 by $k = 13$), with residual $r_2 2^{-k}$, here 2^{-13} at $k = 13$ —the next trivial zero ($\text{ord}_{s=-1} = r_2 = 1$, distance 2) bleeding in, confirming the signature reading.

hR/w and discriminant. With $H(\varepsilon) = G(\varepsilon) - m \log(1 + \varepsilon)$ and $A = e^{H(-1)}$:

$$\mathbb{Q}(\sqrt{5}) : A = 0.24060591253 = \frac{1}{2} \log\left(\frac{1+\sqrt{5}}{2}\right); \quad \mathbb{Q}(\sqrt{2}) : A = 0.44068679351 = \frac{1}{2} \log(1+\sqrt{2}) \quad (13 \text{ digits}),$$

and $|d_K| = (2^{r_1}(2\pi)^{r_2}A/\text{Res})^2$ returns 5.0 and 8.0 exactly. Thus signature, hR/w , and $|d_K|$ are all recovered from the single expansion about $\beta = 1$.

5 Discussion

The dictionary is a reading, not a theorem of arithmetic, and it is honest to say so: each entry is a corollary of a classical special-value identity, repackaged as a feature of the prime gas’s real-axis thermodynamics. Its value is the *simultaneity*—one Fisher expansion, four invariants, no continuation and no appeal to the nontrivial zeros—and the clean separation it forces between what the real axis can report (distances, orders, special values) and what it cannot (zero locations; the Siegel wall of Remark 1). Further sub-leading coefficients read only higher derivatives of the same special values, of diminishing arithmetic interest. A natural next object, left open here, is the Euler–Kronecker constant γ_K [6] as the constant term’s generalization.

References

- [1] B. L. Julia, *Statistical theory of numbers*, in *Number Theory and Physics*, Springer Proc. in Physics **47** (1990); and *Thermodynamic limit in number theory: Riemann–Beurling gases*, Physica A **203**, 425 (1994).
- [2] D. Spector, *Supersymmetry and the Möbius inversion function*, Commun. Math. Phys. **127**, 239 (1990).
- [3] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Math. (N.S.) **1**, 411 (1995).
- [4] J. Neukirch, *Algebraic Number Theory*, Grundlehren **322**, Springer (1999) (functional equation of Λ_K , Γ -factors, class-number formula).
- [5] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloq. Publ. **53** (2004) (approximate functional equation, Artin L -functions).
- [6] Y. Ihara, *On the Euler–Kronecker constants of global fields and primes with small norms*, in *Algebraic Geometry and Number Theory*, Progr. Math. **253**, Birkhäuser (2006).
- [7] IG-PRIMON-T1 receipts (this work): `module_e_radius_finding.py`, `T1.lemma_E1_proof_v0.2.md`, `T1.adversarial_audit_v0.1.md`.